Pass-Through, Welfare, and Incidence under Product Returns in Perfect Competition

Draft notes

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1 Introduction

Who bears a tax, and how much of it passes through into the price, is the question of tax incidence (Weyl and Fabinger, 2013). A sales tax and a platform's referral fee are examples. The pass-through and incidence of such taxes are well studied when purchases are final. Online purchases, however, are often returned. Retailers expect about one in five online sales to be returned.[1] The seller receives the price only when the buyer keeps the good, so every unit sold bears not only its own tax but also the costs of returned units. Those costs include the production cost, the return handling cost, and any tax or fee that is not fully recovered. This extra cost can make pass-through stronger. The price, however, also affects the return rate. A higher price can raise the return rate, because a higher refund makes returning more attractive, or lower it, when it screens out lower-valued buyers, who return more than the average buyer. The return rate, which can move in either direction with the price, also affects the seller's profit margin, the quantity supplied, and the equilibrium price that clears the market. So it is not clear whether tax pass-through is stronger or weaker with returns, or even negative. The question applies to supply-side shocks more generally, such as the tax, production cost, which is incurred on every sale, and return handling cost, which is incurred only when the buyer returns the product.

In practice, the tax on a sale also depends on whether the product is kept or returned. When a purchase is returned, the sales tax is refunded in full. A platform's referral fee is refunded only in part. Amazon deducts the fee from each sale, 15 percent of the sale price in many categories, and keeps up to 20 percent of it on a return.[2] Whether the seller should recover the tax in full on a return, or the platform should be allowed to keep part of it, is an open question. Platforms defend keeping part of the fee as covering the cost of handling returns, yet that part is kept whenever a paid order is refunded, even if the product never shipped. This paper studies the pass-through, incidence, and welfare effects of the tax in a competitive market, including the effect of a tax charged when the good is returned. Each buyer decides whether to buy and, after learning the fit, whether to keep the good, and each seller decides how much to supply. These supply-side shocks reach the buyer's purchase and return decisions only through the price. They reach the seller directly, as part of the cost of each unit sold, and so shift supply.

[1] National Retail Federation and Happy Returns, *2025 Retail Returns Landscape*, October 15, 2025, `https://nrf.com/research/2025-retail-returns-landscape`. The report estimates that 19.3 percent of online sales will be returned in 2025.

[2] Amazon, *Selling on Amazon pricing*, `https://sell.amazon.com/pricing`, and *North America selling fees*, `https://sell.amazon.com.sg/north-america/pricing`. Amazon calls the amount it keeps the refund administration fee, the lesser of five dollars or 20 percent of the referral fee.

A tax creates a wedge between what the buyer will pay and what the good costs the seller. Without returns, the wedge leads to underprovision on the purchase margin, because some beneficial trades do not happen. Returns add a second margin, the decision to keep the good.

2 Model

We study the pass-through of a tax charged by the platform and of a cost increase, such as a rise in return handling cost or in production cost, in a homogeneous-good market under perfect competition with product returns. The product sells at list price p. Consumers indexed by i differ in return hassle κ_i, the disutility of sending the good back, and in the distribution G_i of post-purchase match quality u. After buying, type i draws $u \sim G_i$. In the baseline model, the refund is restricted to equal the list price, so the consumer's utility is $u - p$ when he keeps the product and $-\kappa_i$ when he returns it.

Individual consumer surplus from buying at price p is

$$cs_i(p) = \mathbb{E}_{u \sim G_i}\big[\max(u - p,\, -\kappa_i)\big].$$

Consumer i's purchase decision can be summarized by willingness to pay v_i, defined as the price solving $cs_i(v_i) = 0$. Type i buys if and only if $p \le v_i$. Across types, v_i has distribution F with density f, so gross demand is $D(p) = 1 - F(p)$.

Consumer i returns the item with probability $\phi_i(p) = \Pr(u < p - \kappa_i)$. Averaging it over consumers with the same willingness to pay v gives $\phi(v, p) \equiv \mathbb{E}\big[\phi_i(p) \;\big|\; v_i = v\big]$. The aggregate return rate $r(p)$ is derived by further taking the average over consumers who make a purchase at p, equal to

$$\frac{1}{D(p)} \int_p^\infty \phi(v, p)\, dF(v).$$

The return rate depends on how consumers purchase and on how each purchaser returns. The price shapes the return rate through both. Therefore, the effect of the price can be decomposed into a behavioral channel and a selection channel,

$$r'(p) = \underbrace{\frac{1}{D(p)} \int_p^\infty \frac{\partial \phi(v, p)}{\partial p}\, dF(v)}_{\text{Behavioral channel}} + \underbrace{\frac{f(p)}{D(p)} \big(r(p) - \phi(p, p)\big)}_{\text{Selection channel}},$$

The behavioral channel is the change in return probability holding the purchase cutoff fixed. A higher price lowers the utility of keeping, so purchasers return more often. The selection channel captures the change in who buys. Its sign is undetermined without additional assumptions. In general, this channel is non-positive because higher willingness to pay lowers the return probability, so buyers at the margin return more often than the average buyer and $\phi(p, p) \ge r(p)$.[3] The two channels then move $r'(p)$ in opposite directions. Below, $r'(p)$ is left unsigned.

Producer j takes the list price p and the return rate r as given and chooses quantity q_j at cost $C_j(q_j)$, with strictly increasing marginal cost. The platform charges a unit tax $t < p$,[4] so the seller receives $p - t$ on a kept unit. On a returned unit the seller refunds the full price, so the buyer pays nothing. The seller pays net handling h on the good, which is gross handling cost minus scrap

[3] As when $u = v_i + \varepsilon$, with ε independent of v_i and return hassle independent of v_i.

[4] We use a unit tax rather than the platform's ad valorem referral fee. An ad valorem tax interacts with the price inside effective marginal revenue even when the return rate is fixed, and the pass-through formula is then hard to interpret at a positive tax.

value, and a share $\beta \in [0,1]$ of the tax is retained by the platform on each returned unit. The seller therefore loses $\beta t + h$ on that returned unit. We maintain $p + h - (1-\beta)t > 0$. This is the seller's loss when a kept sale is returned. With return rate $r(p)$, the seller's expected payoff per unit is the effective marginal revenue

$$\mu(p, h, t, \beta) = \big(1 - r(p)\big)(p - t) - r(p)(\beta t + h). \tag{1}$$

2.1 Welfare measure

Consumer surplus is the sum of individual consumer surplus over buyers who purchase. A tax or a cost shock affects consumer surplus only through the list price. A higher price removes the marginal buyer, who had zero surplus, so the lost purchase does not reduce consumer surplus. For an inframarginal buyer, who still purchases, the utility of keeping falls by the price increase. The utility of returning does not change, because the higher price raises the payment and the refund by the same amount. The return probability rises, but the buyer who switches from keeping to returning was indifferent, so the switch does not reduce consumer surplus. The remaining loss is one unit of utility on each unit that is kept. There are $D(p)(1 - r(p))$ such units, so

$$CS'(p) = -D(p)\big(1 - r(p)\big). \tag{2}$$

Producer surplus is the sum of firm profits $\mu q_j - C_j(q_j)$. A tax or a cost shock affects producer surplus only through the effective marginal revenue, and by the envelope theorem a one-dollar rise in the effective marginal revenue raises a firm's profit by the quantity it chooses. Aggregating quantity supplied over firms gives

$$\frac{dPS}{d\mu} = S(\mu), \tag{3}$$

where $S(\mu)$ is market supply.

3 Pass-through

We study pass-through of a production-cost increase, a handling-cost increase, and the platform's unit tax, and denote these pass-throughs by the subscripts c, h, and t. For the production-cost increase, we consider a parallel increase in marginal production cost, so marginal production cost is $C_j'(q_j) + c$. These three shocks shift supply but not demand, because consumers' purchase and return decisions respond only to the list price p.

Pass-through $\rho_x \equiv dp/dx$, for $x \in \{c, h, t\}$, is determined by the market clearing condition. Its general form is

$$m(p, x) \equiv D(p) - S\big(\mu(p, x), x\big). \tag{4}$$

Market clearing requires $m(p, x) = 0$. Differentiating $m = 0$ gives

$$\rho_x = \frac{-S'(\mu)\,\mu_x - S_x}{S'(\mu)\,\mu_p - D'(p)}. \tag{5}$$

Here μ_p and μ_x are partial derivatives of $\mu(p, x)$, and S_x is the direct shift of supply at a fixed effective marginal revenue. The production-cost shock does not enter μ, so $\mu_c = 0$, yet a unit increase in c shifts supply as if μ fell by one unit, so $S_c = -S'(\mu)$. The handling cost and the tax have no direct effect on supply at a fixed effective marginal revenue, so $S_h = S_t = 0$. The handling cost reduces the effective marginal revenue when a consumer returns the product, so $\mu_h = -r(p)$.

The tax reduces effective marginal revenue by 1 on a kept unit and by β on a returned unit, so $\mu_t = -\theta$ with $\theta \equiv 1 - r(p) + \beta r(p)$. This factor is positive because $\theta = 1 - r + \beta r \geq 1 - r > 0$. Substituting these into (5) gives the three pass-throughs,

$$\rho_c = \frac{S'(\mu)}{S'(\mu)\,\mu_p - D'(p)}, \qquad \rho_h = r(p)\,\rho_c, \qquad \rho_t = \theta\,\rho_c, \tag{6}$$

In the classical model without returns, a unit tax and a unit production-cost increase create the same wedge, so $\rho_t = \rho_c$. With returns, each shock shifts the supply curve to the left by $S'(\mu)$ times a scale factor. The factor is 1 for production cost, r for handling cost, and θ for the tax. Since supply and demand are locally linear, pass-through depends only on how far the supply curve shifts. That shift is proportional to the factor, so each pass-through is ρ_c scaled by its factor.

These pass-throughs take a more transparent form in price elasticities, which also exposes the role of returns. The demand and supply elasticities are

$$\varepsilon_D(p) \equiv \frac{pD'(p)}{D(p)} < 0, \qquad \varepsilon_S(p) \equiv S'(\mu)\,\mu_p\,\frac{p}{S(\mu)}.$$

The supply elasticity ε_S has the sign of μ_p, which can be written explicitly as

$$\mu_p = (1 - r) - r'\big[p + h - (1 - \beta)t\big]. \tag{7}$$

With no returns, $r = 0$ and $r' = 0$, so $\mu_p = 1$ and the supply elasticity is positive. With returns, the supply elasticity can be negative. A higher price raises the return rate. When the behavioral channel is strong enough, the higher price can reduce effective marginal revenue and thus lower quantity supplied.

Multiplying numerator and denominator of ρ_c by $p/S(\mu)$ gives the elasticity form,

$$\rho_c = \frac{\varepsilon_S(p)/\mu_p}{-\varepsilon_D(p) + \varepsilon_S(p)}. \tag{8}$$

The pass-through ρ_c is the ratio of the excess-demand gap a cost increase opens to the rate at which a price change closes it. A production-cost increase reduces supply by $S'(\mu)$ and opens excess demand of that size. An increase in the list price closes that gap at rate $S'(\mu)\,\mu_p - D'(p)$. Compared to the classical no-return formula, which sets $\mu_p = 1$, the returns formula scales the pass-through by $1/\mu_p$. When μ_p is above unity, one dollar of list price restores more than one dollar of effective marginal revenue, so a smaller price movement closes the gap. When $0 < \mu_p < 1$, the return rate rises in price and one dollar of list price restores less than one dollar of effective marginal revenue, so the price must move more and pass-through is larger than the classical value. When μ_p is negative, the effective marginal revenue decreases in the list price, and the sign of pass-through depends on the magnitudes of the elasticities. In particular, negative pass-through can happen when supply is more elastic than demand in absolute value, so lowering the price raises supply by more than it raises demand and clears the excess demand. This is in sharp contrast to the model without returns, where pass-through is always positive. The following two results summarize the findings.

Result 1 (Negative pass-through)**.** *We have $\rho_c < 0$ if and only if $\varepsilon_S < 0$ ($\mu_p < 0$) and $|\varepsilon_S| > -\varepsilon_D$.*

Result 2. *Relative to the correct returns-adjusted ρ_c, the classical no-return pass-through formula overestimates pass-through when $\mu_p > 1$, agrees with it when $\mu_p = 1$, and underestimates it when $0 \leq \mu_p < 1$.*

Result 2 depends on which channel dominates the return rate, on the net handling cost, and on the share β of the tax kept on a return. When the selection channel dominates, $r'(p) < 0$. A higher price lowers the return rate and saves cost, so μ_p rises above one. When the behavioral channel dominates, $r'(p) > 0$. A higher price raises the return rate and adds cost, so μ_p falls below one. A larger handling cost and a low refund of the tax amplify both effects. A low refund leaves the seller with less of the tax on a return, so the return is more costly.

4 Local incidence

Having established how prices respond to different shocks, we turn to local incidence, defined as the ratio of the change in consumer surplus to the change in producer surplus for a small shock x. Before a discussion of the incidence, it is useful to see the pass-through under different primitives and to show consumer surplus and producer surplus on the same graph.

Figure 1 shows demand, the inverse supply curve, consumer surplus, producer surplus, and how the inverse supply curve moves after a production-cost increase. The inverse supply curve gives the price required to supply a given quantity. Consumer surplus is the area between gross demand $D(p)$ and returns $D(p)r(p)$ above p. Producer surplus is the area to the left of the supply curve and below p. For a unit production-cost increase, quantity supplied decreases at a fixed price, and the reduction is the same as a one-unit fall in the effective marginal revenue.

The left panel shows the case where the selection channel is stronger than the behavioral channel, so that $r(p)$ falls with p. This can be seen from the graph as returns $D(p)r(p)$ taking a smaller share of gross demand $D(p)$ at a higher p. A higher p screens out consumers with a high return probability, so the effective marginal revenue improves and the seller supplies more, as illustrated by the locally upward-sloping supply curve. On that region, a unit production-cost increase shifts the inverse supply curve inward at a given quantity, so the equilibrium price rises.

The right panel shows the case where the behavioral channel is stronger than the selection channel, so that $r(p)$ rises with p. This can be seen from the graph as returns $D(p)r(p)$ taking a larger share of gross demand $D(p)$ at a higher p. The inverse supply curve is hump-shaped. It is not profitable to supply at a high p, because the return rate is so high that almost any sale leads to a return. It is also not profitable to supply at a low p, because the low profit margin is not enough to cover the production cost. A higher price raises the return probability for purchasers. The seller loses the price and the handling cost on the additional returns. When this loss outweighs the extra revenue on kept sales, the effective marginal revenue falls and the seller supplies less, as illustrated by the locally downward-sloping supply curve. On that region, a unit production-cost increase shifts the inverse supply curve inward at a given quantity, so the equilibrium price falls.

One observation is that the incidence is the same for the three shocks. Intuitively, at a fixed price each shock reduces quantity supplied as if the effective marginal revenue had fallen. The three falls differ only in magnitude. Supply and demand are locally linear, so pass-through and the quantity change depend only on that magnitude. The changes in consumer surplus and producer surplus are both locally linear in how far the supply curve shifts, and that magnitude cancels in the ratio. Mathematically, let $Q \equiv D(p)$ and $r \equiv r(p)$ denote the equilibrium gross quantity and the return rate. A shock x affects consumer surplus only through the price, so $\frac{dCS}{dx} = -Q(1-r)\rho_x$. This change scales with ρ_x and hence with the local supply shift. Producer surplus, written as a function of x, is

$$PS(x) = \int_0^{\mu(p(x),x)} S(z,x)\,dz.$$

A shock x affects producer surplus through pass-through, through the direct effect on the effective

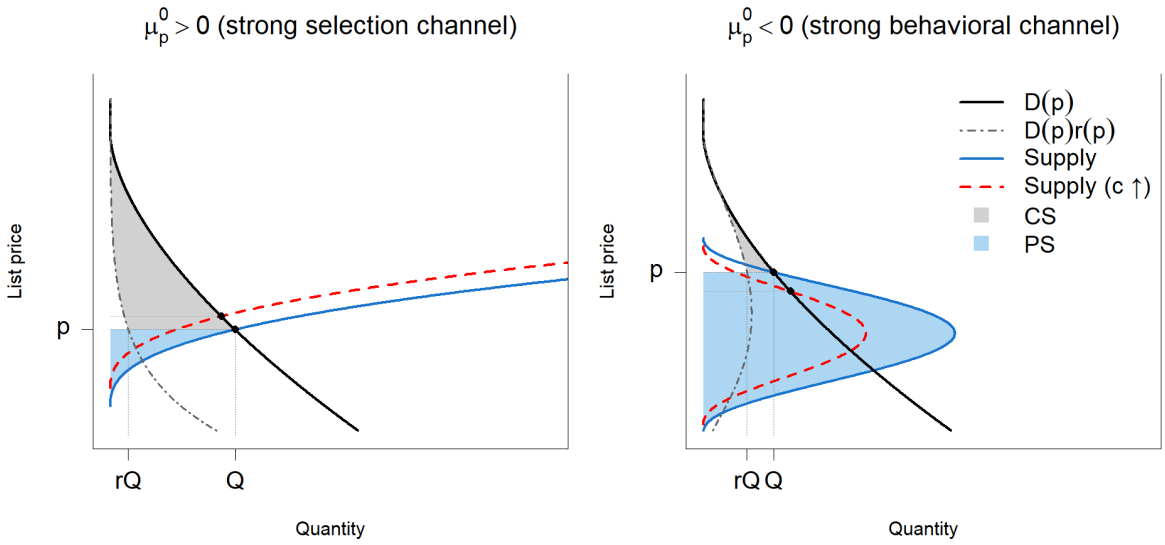


Figure 1: Consumer surplus, producer surplus, and supply after a production-cost increase.

marginal revenue, holding the price fixed, and through a shift of the supply curve, holding the price and effective marginal revenue fixed. Differentiating in x and using $S\big(\mu(p(x),x),x\big) = Q$ gives $dPS/dx = Q\,d\mu(p(x),x)/dx + \Sigma_x$, where $\Sigma_x \equiv \int_0^{\mu(p(x),x)} (\partial S(z,x)/\partial x)\,dz$. Because $d\mu(p(x),x)/dx = \mu_p \rho_x + \mu_x$,

$$\frac{dPS}{dx} = Q\mu_p\rho_x + Q\mu_x + \Sigma_x. \tag{9}$$

Table 1 shows that the joint effect of these three channels is also proportional to pass-through and to the degree of the local supply shift.

Shock x	ρ_x	μ_x	Σ_x	dCS/dx	dPS/dx	I_x
Production cost c	ρ_c	0	$-Q$	$-Q(1-r)\rho_c$	$Q(\mu_p\rho_c - 1)$	$\dfrac{(1-r)\rho_c}{1-\mu_p\rho_c}$
Handling cost h	$r\rho_c$	$-r$	0	$-Q(1-r)r\rho_c$	$Qr(\mu_p\rho_c - 1)$	$\dfrac{(1-r)\rho_c}{1-\mu_p\rho_c}$
Tax t	$\theta\rho_c$	$-\theta$	0	$-Q(1-r)\theta\rho_c$	$Q\theta(\mu_p\rho_c - 1)$	$\dfrac{(1-r)\rho_c}{1-\mu_p\rho_c}$

Table 1: Local incidence for each shock.

The local incidence ratio is therefore the same for the three shocks. The next result formalizes this and shows that the common ratio is positive.

Result 3. *For production cost, handling cost, and the tax, the local incidence ratio is identical and positive.*

$$I_x = \frac{(1-r)\rho_c}{1-\mu_p\rho_c} > 0. \tag{10}$$

Proof. Since $1 - r > 0$, the ratio is positive if ρ_c has the same sign as $1 - \mu_p\rho_c$. Multiplying the pass-through formula (8) by μ_p gives $\mu_p\rho_c = \varepsilon_S/(\varepsilon_S - \varepsilon_D)$. This exceeds one if and only if $\varepsilon_S < 0$ and $|\varepsilon_S| > -\varepsilon_D$. Result 1 states that this is the necessary and sufficient condition for $\rho_c < 0$. □

For a unit increase in production cost, the numerator $(1-r)\rho_c$ is the per-unit consumer loss on kept units. The denominator $1-\mu_p\rho_c$ is the per-unit producer loss. Before the price adjusts, the direct effect through the supply curve shift lowers producer surplus by Q. The indirect price pass-through effect moves the effective marginal revenue by $\mu_p\rho_c$ per unit, and the envelope $dPS/d\mu = Q$ restores producer surplus by $Q\mu_p\rho_c$. Together, the net change in producer surplus is $Q(\mu_p\rho_c - 1)$.

The incidence is positive even when pass-through is negative because the recovery from the indirect price pass-through effect more than offsets the direct loss from the supply curve shift. With negative pass-through, the new equilibrium price is lower and quantity demanded is higher. Market clearing requires the new equilibrium quantity supplied to exceed the original quantity. The price channel must therefore raise quantity supplied by more than the initial decrease caused by the inward shift of the supply curve. This implies that the price channel raises the effective marginal revenue by more than one unit, so the denominator is negative when pass-through is negative. The reasoning in the opposite direction holds for positive pass-through.

The result has a striking implication that a production cost increase can be Pareto-improving. The uniformly positive incidence ratio means consumer and producer surplus always move together, regardless of the sign of pass-through. When pass-through is negative, raising production cost, handling cost, or the tax lowers the price, reduces return rates, and saves enough handling costs to more than offset the direct resource drain. Both consumers and producers benefit. When pass-through is positive, all three cost shocks reduce both surpluses, as in the classical model, where $\mu_p = 1$ and $\rho_c > 0$.

4.1 First-order effect on joint surplus and ex-ante ex-post decomposition

Joint surplus is $W \equiv CS + PS$. A unit increase in production cost changes it by the sum of the consumer and producer changes in Table 1,

$$\frac{dW}{dc} = -Q(1-r)\rho_c + Q(\mu_p\rho_c - 1). \tag{11}$$

Substituting μ_p into (11) gives

$$\frac{dW}{dc} = -Q\Lambda, \qquad \Lambda \equiv 1 + r'(p)\rho_c\big[p + h - (1-\beta)t\big]. \tag{12}$$

Pass-through of the production cost changes the price by ρ_c on each unit. Without returns, buyers lose ρ_c on each of the Q units sold, so they lose $Q\rho_c$. Sellers gain ρ_c on each of those units, so they gain the same amount. These price effects cancel, so what remains is the direct effect of the higher production cost on the equilibrium quantity sold, a change of $-Q$.

With returns, consider the case in which the price does not change who buys or who returns. Buyers lose ρ_c on each kept unit and lose nothing on a returned unit, because the refund equals the price. There are $Q(1-r)$ kept units, so buyers lose $Q(1-r)\rho_c$. Sellers gain the same amount on those units. These price effects still cancel, and the direct loss $-Q$ remains.

A change in who buys or who returns does not add a further consumer loss. A buyer who stops buying had zero surplus. The higher price lowers the value of keeping, but the buyer who switches from keep to return was indifferent between the two, so the switch offsets that lower value. The cost of the switch falls on the seller, who gives up the price p, pays the handling cost h, and recovers the refunded part $(1-\beta)t$ of the tax. The price moves the return rate by $r'(p)\rho_c$, so these switches change joint surplus by $-Qr'(p)\rho_c\big[p + h - (1-\beta)t\big]$. Added to the direct loss $-Q$, the change is $-Q\Lambda$.

Handling cost scales this change by r, and the tax scales it by θ.

$$\frac{dW}{dh} = -Qr\Lambda, \qquad \frac{dW}{dt} = -Q\theta\Lambda. \tag{13}$$

Freezing the return rate separates the ex-ante purchase margin from the ex-post keep-or-return margin. With $r'(p) = 0$, the factor Λ equals one. A unit production-cost increase then shifts supply up by one dollar at every quantity. The price changes in consumer and producer surplus cancel. Joint surplus then falls by the extra dollar of production cost on each of the Q units sold, a rectangle of height one and width Q. Handling cost leaves the rectangle Qr, and the tax leaves the rectangle $Q\theta$. These rectangles are the ex-ante term, what the change would be if returns did not respond to the price. The gap between the actual change $-Q\Lambda$ and this rectangle is the ex-post term $-Q\,r'(p)\rho_c\big[p + h - (1-\beta)t\big]$, scaled by r or by θ for the other two shocks.

Result 4 (Ex-ante and ex-post decomposition)**.**

$$\frac{dW}{dc} = -Q - Q\,r'(p)\rho_c\big[p + h - (1-\beta)t\big], \tag{14}$$

$$\frac{dW}{dh} = -Qr - Qr\,r'(p)\rho_c\big[p + h - (1-\beta)t\big], \tag{15}$$

$$\frac{dW}{dt} = -Q\theta - Q\theta\,r'(p)\rho_c\big[p + h - (1-\beta)t\big]. \tag{16}$$

The ex-post margin offsets the ex-ante loss when $0 < \Lambda < 1$, amplifies it when $\Lambda > 1$, and reverses it when $\Lambda < 0$. Because $dW/dc = -Q\Lambda$, joint surplus falls when Λ is positive and rises when Λ is negative. Result 3 says that the two surpluses move together. They fall when $\rho_c > 0$ and rise when $\rho_c < 0$, so Λ has the same sign as ρ_c.

When pass-through is positive, $r'(p)$ decides whether the ex-post margin offsets or amplifies the ex-ante loss. When the selection channel dominates, $r'(p) < 0$, so $\mu_p > 0$. The higher list price screens out frequent returners, the return rate falls, and the effective marginal revenue improves. The ex-post margin is then a gain, and it cuts the loss from Q to $Q\Lambda$ with $0 < \Lambda < 1$. When the behavioral channel dominates, $r'(p) > 0$, so $\mu_p < 1$. The higher list price makes keeping less attractive, the return rate rises, and the effective marginal revenue worsens. The ex-post margin is then a further loss, and it raises the loss from Q to $Q\Lambda$ with $\Lambda > 1$. In both cases the gap from the ex-ante loss is larger when the return rate responds more strongly to the price, or when a return is more costly to handle.

When pass-through is negative, the ex-post margin reverses the ex-ante loss and joint surplus rises. Result 1 says that negative pass-through occurs only when the behavioral channel dominates. In this case, a lower price makes keeping more attractive, so the return rate falls and the seller pays the return cost on fewer units. When that saving outweighs the lower revenue on kept units, effective marginal revenue rises as the price falls, and supply slopes down. The behavioral effect is so strong that a lower price raises supply by more than demand and closes the excess-demand gap. The ex-post margin is then a gain, and it turns the loss of Q into a gain of $Q|\Lambda|$. The gain is larger than the ex-ante loss when the return rate responds strongly to the price, when the return cost is high (large β and h), or when the magnitude of pass-through is large.

The following result summarizes these three cases by pass-through and $r'(p)$. It shows the direction of the error when the no-return formula is used.

Result 5 (No-return joint-surplus bias)**.** *Consider a unit production-cost increase. Relative to the returns-adjusted change $dW/dc = -Q\Lambda$, the ex-ante formula $dW/dc = -Q$*

(i) overestimates the joint-surplus loss if and only if $r'(p) < 0$, in which case $0 < \Lambda < 1$;

(ii) underestimates the joint-surplus loss if and only if $r'(p) > 0$ and $\rho_c > 0$, in which case $\Lambda > 1$;

(iii) overestimates the joint-surplus loss with the wrong sign if and only if $r'(p) > 0$ and $\rho_c < 0$. It reports a loss of Q when joint surplus rises by $Q|\Lambda|$.

For a handling-cost shock the same cuts compare $-Qr\Lambda$ with $-Qr$. For the tax they compare $-Q\theta\Lambda$ with $-Q\theta$.

Proof. The three regimes above give the direct statements. For the converses, $0 < \Lambda < 1$ forces $\rho_c > 0$ by positive incidence and $r'(p)\rho_c < 0$, so $r'(p) < 0$. This is (i). $\Lambda > 1$ forces $r'(p)\rho_c > 0$ and $\rho_c > 0$, so $r'(p) > 0$. This is (ii). $\Lambda < 0$ forces $\rho_c < 0$ and therefore $r'(p) > 0$. This is (iii). Since r and θ are positive, the handling-cost and tax comparisons follow the same logic. □

The sign of the ex-post margin depends on the relative strength of the two channels, and it is not monotone. When the selection channel is stronger, supply still slopes up and pass-through is positive. The higher price screens out consumers with a high return probability, the return rate falls, and the ex-post margin is a gain. When the behavioral channel dominates but supply is not yet both downward sloping and more elastic than demand in absolute value, pass-through remains positive. The higher price raises the return probability of the remaining consumers, the return rate rises, and the ex-post margin is a loss. When the behavioral channel is strong enough that supply is both downward sloping and more elastic than demand in absolute value, pass-through is negative. The lower price lowers the return probability of the remaining consumers, the return rate falls, and the ex-post margin is again a gain. Result 6 states these three cases as thresholds on $r'(p)$.

Result 6 (Thresholds for the ex-post term)**.** *Let*

$$r'_0 \equiv \frac{1 - r(p)}{p + h - (1 - \beta)t}, \qquad \bar{r}' \equiv \frac{1 - r(p) - D'(p)/S'(\mu)}{p + h - (1 - \beta)t}. \tag{17}$$

Then $0 < r'_0 < \bar{r}'$. For the ex-post term $-Qr'(p)\rho_c\big[p + h - (1 - \beta)t\big]$,

(i) if $r'(p) < 0$, supply slopes up, $\rho_c > 0$, and the term is a gain;

(ii) if $0 < r'(p) < \bar{r}'$, then $\rho_c > 0$ and the term is a loss. Supply slopes up when $r'(p) < r'_0$ and slopes down when $r'_0 < r'(p) < \bar{r}'$;

(iii) if $r'(p) > \bar{r}'$, supply slopes down and is more elastic than demand in absolute value, $\rho_c < 0$, and the term is a gain.

Proof. From (7), $\mu_p = (1-r) - r'\big[p+h-(1-\beta)t\big] = (r'_0 - r')\big[p+h-(1-\beta)t\big]$. The supply elasticity has the sign of μ_p, so supply slopes up when $r' < r'_0$ and slopes down when $r' > r'_0$. Supply slopes down and is more elastic than demand in absolute value when $\varepsilon_S < \varepsilon_D < 0$. The demand elasticity is negative. At the equilibrium $S = D$, the gap $\varepsilon_S - \varepsilon_D$ has the sign of $S'\mu_p - D'$, the denominator of ρ_c in (6). Substituting μ_p gives $S'\big[p + h - (1 - \beta)t\big](\bar{r}' - r')$. This factor is positive, so the gap and ρ_c are negative when $r' > \bar{r}'$. □

For the tax, platform revenue is part of social surplus. The platform receives t on a kept sale and βt on a return, so revenue is $G = Qt\theta$. Social surplus is $V \equiv W + G$. The marginal effect of the tax on social surplus is

$$\frac{dV}{dt} = -Q\theta\Lambda + Q\theta + t\theta\rho_c\big[D'(p)\,\theta - Q(1 - \beta)r'(p)\big]. \tag{18}$$

Without returns, $r'(p) = 0$ and $\theta = 1$. The ex-ante term $-Q$ is a transfer to the platform, and there is no ex-post term. What remains is the tax term $t\,dQ/dt$, the underprovision caused by the wedge between the consumer's value and the production cost. With returns, the ex-ante term $-Q\theta$ is still a transfer to the platform. What remains is the ex-post term and the tax term. Expanding the tax term and cancelling the refunded tax terms in $(1-\beta)$, the remaining terms are

$$-Q\theta\, r'(p)\rho_c(p+h) + D'(p)\, t\theta^2\rho_c.^5$$

The first term is the ex-post margin net of the refunded tax $(1-\beta)t$, which is a pure transfer between seller and platform and leaves only the real return cost $p+h$. Since $Q, p+h > 0$, its sign depends only on the return-rate change $r'(p)\rho_t = r'(p)\,\theta\rho_c$. The second term is the tax wedge on the ex-ante purchase margin. The tax on existing units is a transfer to the platform, and on marginal units the buyer's indifference at p and the seller's effective marginal revenue equaling marginal cost leave private surplus unchanged, so only the expected tax $t\theta$ per unit enters social surplus. Multiplying $t\theta$ by the quantity change $D'(p)\rho_t = D'(p)\,\theta\rho_c$ gives $D'(p)\,t\theta^2\rho_c$, which is negative when pass-through is positive and positive when pass-through is negative.

By Result 6, the two margins depend on the strength of the behavioral channel relative to the selection channel. When the behavioral channel is weak, pass-through is positive and quantity falls, so the ex-ante margin is a loss, while the higher price screens out high-return buyers and makes the ex-post margin a gain. The two margins trade off. When the behavioral channel is intermediate, pass-through stays positive and quantity falls, while the return rate rises with the price, so both margins are losses. When the behavioral channel is very strong, pass-through turns negative, so the price and return rate fall while quantity rises, and both margins are gains.

When $t > 0$ and $r > 0$, raising the retained tax share β lowers effective marginal revenue on returns and shifts supply inward like a tax increase, while the extra tax kept on existing returns is a transfer to the platform. Consequently, β moves social surplus in the same direction as the tax.

Result 7 (Retained tax share and social surplus)**.** *Suppose $t > 0$ and $r(p) > 0$. Then*

$$\frac{dV}{d\beta} = \frac{rt}{\theta}\,\frac{dV}{dt} = rt\rho_c\big[-Qr'(p)(p+h) + D'(p)\,t\theta\big]. \tag{19}$$

By Result 6:

(i) if $r'(p) > \bar{r}'$, then $dV/d\beta > 0$ (a higher β raises social surplus);

(ii) if $0 < r'(p) < \bar{r}'$, then $dV/d\beta < 0$ (a lower β raises social surplus);

(iii) if $r'(p) < 0$, the ex-post gain and ex-ante loss trade off, so $dV/d\beta$ has the sign of $-Qr'(p)(p+h) + D'(p)\,t\theta$.

Proof. By (1), $\mu_\beta = -rt = (rt/\theta)\mu_t$ and $S_\beta = S_t = 0$. By (5) and (6), $\rho_\beta = (rt/\theta)\rho_t = rt\rho_c$. Because CS and PS depend on (t,β) only through (p,μ), scaling (μ_t, ρ_t) by rt/θ gives $dW/d\beta = (rt/\theta)(dW/dt)$. For revenue $G = Qt\theta$, since $dG/dt = Q\theta + t\,\partial(Q\theta)/\partial p \cdot \rho_t$, $\partial\theta/\partial\beta = r$, and $\rho_\beta = (rt/\theta)\rho_t$,

$$\frac{dG}{d\beta} = Qrt + t\frac{\partial(Q\theta)}{\partial p}\,\rho_\beta = \frac{rt}{\theta}\left(Q\theta + t\frac{\partial(Q\theta)}{\partial p}\,\rho_t\right) = \frac{rt}{\theta}\,\frac{dG}{dt}.$$

[5]The tax term is $t\,d(Q\theta)/dt$. The product rule gives $d(Q\theta)/dt = Q\,d\theta/dt + \theta\,dQ/dt$. Quantity follows demand, so $dQ/dt = D'(p)\rho_t$. The tax weight is $\theta = 1-(1-\beta)r$, and the tax moves the return rate through the price, so $d\theta/dt = -(1-\beta)r'(p)\rho_t$. Substituting both gives $d(Q\theta)/dt = \rho_t[D'(p)\,\theta - Q(1-\beta)r'(p)]$. Added to the ex-post term, the pieces in $(1-\beta)$ cancel. A returned sale pays the platform only part of the tax. The seller recovers the rest, and the platform loses the same amount. These are the same dollars, so they cancel in social surplus. What remains is the real cost of a return.

Adding $dW/d\beta$ and $dG/d\beta$ gives $dV/d\beta = (rt/\theta)\, dV/dt$, and substituting $dV/dt = \theta\rho_c\big[-Qr'(p)(p+h) + D'(p)\, t\theta\big]$ yields (19). Since $t, r, \theta, Q, p+h > 0$ and $D'(p) < 0$, Result 6 implies that in (i) $r'(p) > 0$ and $\rho_c < 0$, so both terms in (19) are positive; in (ii) $r'(p) > 0$ and $\rho_c > 0$, so both terms are negative; and in (iii) $r'(p) < 0$ and $\rho_c > 0$, so $-Qrt\, r'(p)\rho_c(p+h) > 0$ while $D'(p)\, rt^2\theta\rho_c < 0$, leaving the sign of $-Qr'(p)(p+h) + D'(p)\, t\theta$. □

5 Global Incidence

Each row of Table 1 is the local derivative at the current equilibrium, using the current μ_p and ρ_c. A finite change in production cost, handling cost, or the tax integrates that row, with the row re-evaluated at the current equilibrium.

Along such a path, the global incidence ratio $I_{x_0}^{x_1} \equiv \Delta CS/\Delta PS$, when $\Delta PS \neq 0$, is the weighted average of the local ratio implied by $dCS/dx = I(x)\, dPS/dx$,

$$I_{x_0}^{x_1} = \int_{x_0}^{x_1} I(x)\, \omega(x)\, dx, \tag{20}$$

where $\omega(x) \equiv (dPS/dx)/\Delta PS$.

The global ratio can be negative, even though $I(x)$ is positive at every point on the path. If dPS/dx changes sign along the path, the weights $\omega(x)$ take both signs, and the weighted average of positive local ratios can be negative. A finite change therefore need not preserve the sign of local incidence.

A Welfare bias in primitives

Result 5 has an equivalent statement in elasticities. Recall $\varepsilon_r(p) \equiv p\,r'(p)/r(p)$ and $\mu_p = (1 - r(p)) - r'(p)\big[p + h - (1-\beta)t\big]$. Then $r'(p)\big[p + h - (1-\beta)t\big] = r(p)\,\varepsilon_r(p)\,\big[p + h - (1-\beta)t\big]/p$, so

$$\mu_p > 0 \iff \varepsilon_r(p) < \frac{p}{p + h - (1-\beta)t}\,\frac{1 - r(p)}{r(p)}, \tag{21}$$

In these elasticities, Result 5 reads as follows.

(i) overestimation iff $\varepsilon_r(p) < 0$;

(ii) underestimation iff $\varepsilon_r(p) > 0$ and either $\mu_p \geq 0$, or both $\mu_p < 0$ and $|\varepsilon_S| < -\varepsilon_D$;

(iii) overestimation with wrong sign (predicted loss, actual gain) iff $\mu_p < 0$ and $|\varepsilon_S| > -\varepsilon_D$.

Proof. By Result 5, overestimation with $0 < \Lambda < 1$ is $r' < 0$, underestimation is $r' > 0$ with $\rho_c > 0$, and overestimation with wrong sign is $r' > 0$ with $\rho_c < 0$. Since ε_r has the same sign as r', write these in ε_r. For (i), $\varepsilon_r < 0$ forces $\mu_p > 1 - r > 0$ and hence $\rho_c > 0$, so $0 < \Lambda < 1$. When $\mu_p = 0$ and $r(p) < 1$, the derivative $r'(p)$ is positive, $\varepsilon_S = 0$, and $\rho_c = S'(\mu)/(-D'(p)) > 0$, so that case is in (ii). Result 1 translates the sign of ρ_c into conditions on μ_p and $|\varepsilon_S| \gtrless -\varepsilon_D$: with $\varepsilon_r > 0$ this is (ii), and $\rho_c < 0$ is (iii). □

References

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